

The empirical recurrence for OEIS A231250, recovered from its tail

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Abstract

OEIS A231250 records “Empirical recurrence of order 62” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 264$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 64$. Its last coefficient is $c_{62} = 12634030080 \neq 0$, so no further tail satisfies anything shorter; any order-62 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A231250 is “Number of $5 \times n$ 0..3 arrays $x(i,j)$ with each element horizontally, diagonally or antidiagonally next to at least one element with value $(x(i,j)+1) \bmod 4$, no adjacent elements equal, and upper left element zero.”. It has offset 1 and begins

0, 68, 3820, 76136, 1872136, 52280836, 1450042214, 39779238784, ...

The entry states:

Empirical recurrence of order 62 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 06:20:27, revision 7) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 264$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 264$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 62: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 62.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 528$ exact terms from $n = 2$ on, it returns order 62 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{62} c_i a(n-i),$$

c_1	31	c_{17}	−527296537730	c_{33}	−2110705252750477	c_{49}	−240487476432320
c_2	−154	c_{18}	−960773257465	c_{34}	−3050526954734983	c_{50}	−316962149904192
c_3	1738	c_{19}	−313604451426	c_{35}	−1624526466760395	c_{51}	−286435363855104
c_4	−2633	c_{20}	5357986012150	c_{36}	755245096723702	c_{52}	−34619286851360
c_5	−56618	c_{21}	13950632095967	c_{37}	3049590172879841	c_{53}	14569058731904
c_6	−142192	c_{22}	22870811572985	c_{38}	3148635207028453	c_{54}	17710159600640
c_7	−2747769	c_{23}	36469536470951	c_{39}	2070891147363886	c_{55}	23458862589440
c_8	−6714218	c_{24}	−15644246778797	c_{40}	356052884512682	c_{56}	714161742336
c_9	18457595	c_{25}	−183304037168931	c_{41}	−1537515845681688	c_{57}	3555779347456
c_{10}	153848111	c_{26}	−324917007084488	c_{42}	−2311492294321958	c_{58}	212383410176
c_{11}	1196798235	c_{27}	−395751914052846	c_{43}	−2694945928315456	c_{59}	−131682373632
c_{12}	4405903697	c_{28}	−107608628455867	c_{44}	−1302086484379536	c_{60}	94068301824
c_{13}	5644440759	c_{29}	943799938608009	c_{45}	394307343449524	c_{61}	−125337010176
c_{14}	−3107714403	c_{30}	1839353238326358	c_{46}	1115364504179936	c_{62}	12634030080
c_{15}	−49178534573	c_{31}	1753043640973569	c_{47}	1301391730559464		
c_{16}	−217728871749	c_{32}	401601921520534	c_{48}	330323330625768		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 64$, and it is the recurrence OEIS A231250 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{62} - c_1 x^{62-1} - \dots - c_{62}$. Its constant term is $-c_{62} = -12634030080$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component

of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 62 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 62, so $\deg q = 62$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 62, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 16 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 62, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 12634030080.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-62 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A231250.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)
- [5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.